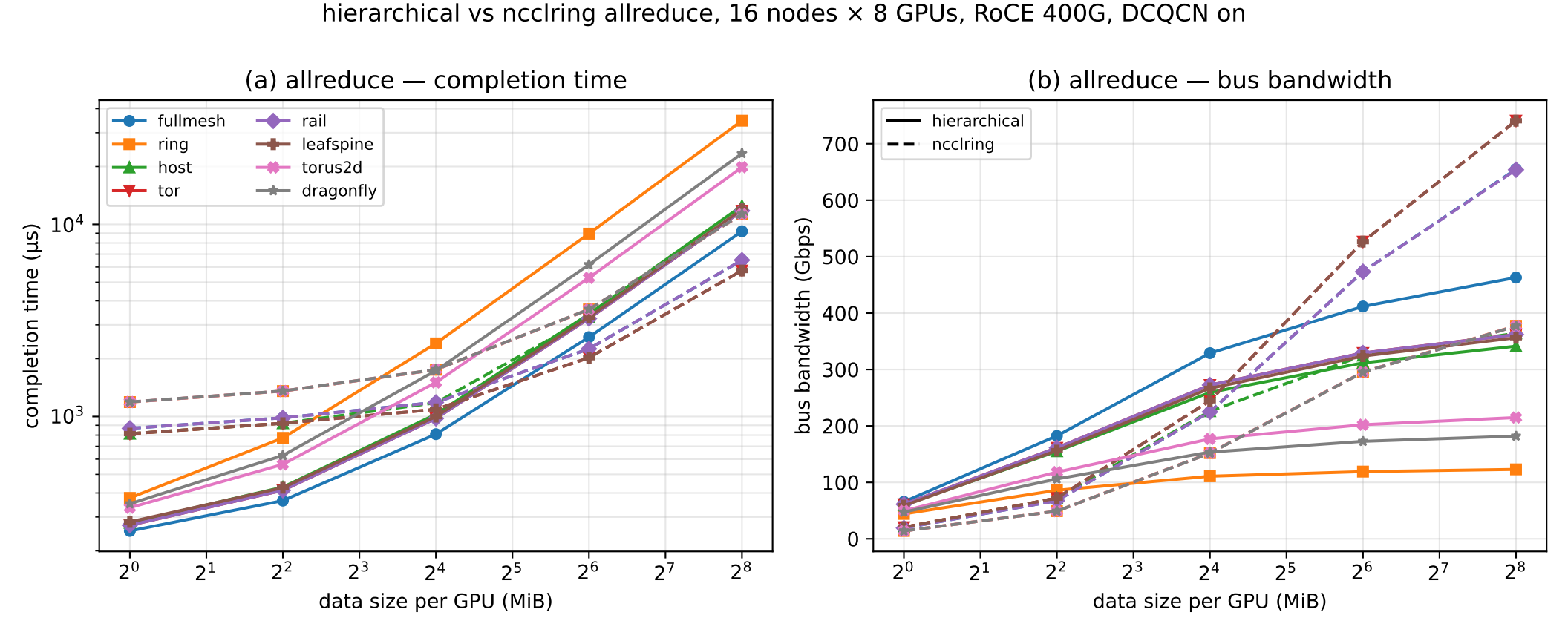
hierarchical vs ncclring allreduce, 16 nodes × 8 GPUs, RoCE 400G, DCQCN on
(a) allreduce — completion time
completion time (µs)
data size per GPU (MiB)
fullmesh
ring
host
tor
rail
leafspine
torus2d
dragonfly
(b) allreduce — bus bandwidth
bus bandwidth (Gbps)
data size per GPU (MiB)
hierarchical
ncclring